arXiv:2609.15930v3 [math.AG] 25 Sep 2026

TWO PROOFS OF THE CASSELS–SWINNERTON-DYER CONJECTURE FOR CUBIC SURFACES

VALERY ALEXEEV AND STEFAN SCHREIEDER

Abstract. The Cassels–Swinnerton-Dyer conjecture asserts that a cubic hypersurface contains a rational point if and only if it contains a point of degree coprime to 3, or, equivalently, a zero-cycle of degree 1. The case of smooth cubic surfaces in characteristic zero has been reduced by Coray (1976) and Voisin (2026) to the case of points of degree 4. We give two independent proofs of this missing case and use a lifting argument to extend the result to smooth cubic surfaces over arbitrary fields. We further give a separate argument for the case of singular cubic surfaces, extending previous work of Coray over perfect fields. Altogether, this proves the Cassels–Swinnerton-Dyer conjecture for cubic surfaces.

Contents

1. Introduction 1
2. Preliminaries and degree reductions 3
3. The degree-five argument: Pfaffian presentations 4
4. The degree-four argument: nets of quadrics 6
5. From characteristic zero to positive characteristic 12
6. Removing the smoothness hypothesis 13
AI disclosure 15
Acknowledgments 15
References 16

1. Introduction

Cassels and Swinnerton-Dyer conjectured that a cubic surface with a point of degree coprime to 3, or, equivalently, a zero-cycle of degree one, has a rational point; see also the related problems of Segre [Seg51, p. 2]. The analogous criteria for quadrics and plane cubic curves are due to Springer [Spr52] and Poincaré [Poi01]; see [Cor76a, Proposition 2.3] for a modern treatment of the latter.

Coray formulated the Cassels–Swinnerton-Dyer conjecture for cubics of arbitrary dimension [Cor76a, Conjecture (CS)]. Over perfect fields, he proved the singular surface case [Cor76a, Proposition 8.2] and reduced the case of smooth cubic surfaces to points of degrees 4 and 10 [Cor76a, Theorem 7.1]. Further developments and related results appeared in [Cor76b, CTC79, CTSD12, BN18, CT20, Ma21, RV22, Bal24, Por25].

In a recent breakthrough, Voisin [Voi26] eliminated the degree-10 case in Coray's reduction: she showed that, in characteristic zero, a smooth cubic surface with a point of degree 10 has a rational point or a point of degree 4. This reduces the conjecture for smooth cubic surfaces in characteristic zero to the following statement, which we prove in this paper.

Theorem 1.1. *Let S be a smooth cubic surface over a field k of characteristic zero. If S has a closed point of degree 4, then it has a rational point.*

Date: September 28, 2026.

2020 *Mathematics Subject Classification.* Primary 14G05; Secondary 14C25, 14J26.

Key words and phrases. Cubic surfaces, rational points, zero-cycles, nets of quadrics, Pfaffian presentations, Cassels–Swinnerton-Dyer conjecture.

In Section 5, we combine this result and the aforementioned work of Coray and Voisin with a lifting argument of Ma [Ma21] to deal with smooth cubic surfaces over arbitrary fields. Finally, in Section 6, we handle singular cubic surfaces over arbitrary fields by reducing to smooth cubic surfaces over $k((t))$, extending Coray's result over perfect fields. Together, these arguments give the following solution of the Cassels–Swinnerton-Dyer conjecture for surfaces; see Theorem 6.1 below for a geometric formulation.

Theorem 1.2. *Let $F \in k[x_0, x_1, x_2, x_3]$ be a nonzero homogeneous cubic form over a field k. Then F represents zero nontrivially over k if and only if it represents zero nontrivially over a finite field extension L/k of degree coprime to 3.*

By results of Segre, Manin [Man86] and Kollár [Kol02], smooth cubic surfaces with a rational point are unirational. We therefore obtain the following corollary.

Corollary 1.3. *Let S be a smooth cubic surface over a field k. Then S is unirational if and only if it contains a zero-cycle of degree 1.*

Another application of Theorem 1.2 to the Hasse principle for cubic hypersurfaces is given by Colliot-Thélène and Wittenberg in [CTW26].

In view of [Cor76a, Ma21, Voi26], the key new ingredient in the proof of Theorem 1.2 is Theorem 1.1 above. We give two independent proofs of the latter statement. The first proof, presented in Section 3, uses a reduction of Cassels (see Proposition 2.4 below) to pass from a point of degree 4 to an effective zero-cycle of degree 5. Using work of Buchsbaum–Eisenbud [BE77] and Beauville [Beau00], we show that the surface either has a rational point or admits a linear 6×6 Pfaffian presentation over the ground field. The following result then completes the argument, where we recall that the Pfaffian $\mathrm{Pf}(A)$ of an alternating $2n \times 2n$ matrix A is a homogeneous polynomial of degree n in its entries satisfying $\mathrm{Pf}(A)^2 = \det(A)$.

Theorem 1.4. *Let k be a field of characteristic different from 2. Let $A = (a_{ij})$ be an alternating 6×6 matrix of linear forms $a_{ij} \in k[x_0, \dots, x_3]$ in four variables. Then the cubic form $\mathrm{Pf}(A)$ given by the Pfaffian of A represents zero nontrivially over k.*

The second proof of Theorem 1.1, given in Section 4, works directly with the given closed point P of degree 4 on S. We construct a suitable tangent direction at P, giving rise to a nonreduced subscheme Z of S of length 8 that is contained in the vanishing locus of three linearly independent quadratic forms Q_1, Q_2, Q_3. If the corresponding quadrics have zero-dimensional base locus, then the cubic form F that cuts out S can be written as $F = H_1Q_1 + H_2Q_2 + H_3Q_3$ for linear forms H_i (see Lemma 4.2), and any k-rational point of $H_1 = H_2 = H_3 = 0$ lies on S. If the base locus of the linear series of quadrics through Z is not zero-dimensional, then we show that either S has a rational point or the base locus is a twisted cubic Γ. If Γ lies in S, then S has a rational point because $\Gamma \cong \mathbb{P}^1_k$ (see Lemma 2.3). Otherwise, $\Gamma \cap S$ has degree 9 and contains the subscheme Z of length 8, so the residual intersection has degree 1 and yields a rational point.

Remark 1.5. Theorem 1.2 implies that a smooth del Pezzo surface of degree 3 over an arbitrary field has a rational point if and only if it has index 1, equivalently, a zero-cycle of degree 1. This equivalence was already known for del Pezzo surfaces of degree at least 4; the case of perfect fields is due to Coray [Cor77], degree 4 over arbitrary fields is due to Amer, Brumer and Leep (see e.g. [EKM08, Corollary 18.7]), and the higher degrees can be deduced from [AB18, §6–§10] and the references therein. The statement also holds in degree 1, as every del Pezzo surface of degree 1 has a rational point, namely the base point of its anticanonical pencil. However, the statement fails for del Pezzo surfaces of degree 2, even in characteristic zero, see [KM17] and [CT20, Remarque 4.3]. The analogue of Theorem 1.2 also fails for quartic curves and surfaces, see [Cor76a, Example 2.8] and [BN18], respectively.

The starting points for this paper were two GPT-assisted proofs obtained independently by the authors; see the AI disclosure below for details of the interactions with GPT and the respective contributions.

2. Preliminaries and degree reductions

Throughout this section, the field k is perfect. The degree of a finite k-scheme is its length over k. We use the associated effective zero-cycle when taking residual intersections.

The reductions in Lemmas 2.1 and 2.2 below are classical, see [Cor76a, Propositions 2.2 and 2.3].

Lemma 2.1. *Let $X \subset \mathbb{P}^3_k$ be the zero locus of a nonzero cubic form. If X has an effective zero-cycle of degree one or two, then $X(k) \neq \varnothing$.*

Proof. It suffices to consider a closed point P of degree two. Its two conjugates span a k-line ℓ. If $\ell \subset X$, it supplies a rational point. Otherwise $[\ell \cap X] - [P]$ is an effective zero-cycle of degree one. □

Lemma 2.2. *Let P be a closed point of degree $d \in \{4, 5\}$ on a smooth cubic surface S/k.*

(1) If P is contained in a k-plane, then $S(k) \neq \varnothing$.

(2) If P is not contained in a k-plane, its geometric points are in general linear position: any four of them span $\mathbb{P}^3$.

Proof. For (1), which goes back to Poincaré [Poi01], let Π be a k-plane that contains P. Since $h^0(\Pi, \mathcal{O}_\Pi(2)) = 6 > d$, there is a conic $Q \subset \Pi$ containing P. Write $C = S \cap \Pi$. If C and Q have no common component, then $[C \cap Q] - [P]$ is effective of degree $6 - d \leq 2$, and Lemma 2.1 applies. Otherwise the greatest common divisor of their equations over k has degree one or two. In the first case it defines a k-line on S; in the second, the residual factor of the cubic defines such a line.

For (2), the Galois group acts transitively on the four-element subsets of the geometric points of P. Thus, if one such subset were coplanar, all would be, and the span of P would be contained in a k-plane, a contradiction. □

The next lemma studies the arithmetic of twisted cubics.

Lemma 2.3. *Let $\Gamma \subset \mathbb{P}^3_k$ be a smooth projective curve over a field k, whose base change to $\bar{k}$ is a twisted cubic. Then $\Gamma \cong \mathbb{P}^1_k$ and $\Gamma \subset \mathbb{P}^3_k$ is a twisted cubic.*

Proof. Since $\Gamma_{\bar{k}}$ is a twisted cubic, $\mathcal{O}(1)|_\Gamma$ is a line bundle of degree three. Hence, $\omega_\Gamma \otimes \mathcal{O}(1)$ is a line bundle of degree one and so Γ is rational. Under the identification $\Gamma \cong \mathbb{P}^1_k$, the embedding $\Gamma \subset \mathbb{P}^3_k$ must then be given by the full linear series $|\mathcal{O}_{\mathbb{P}^1_k}(3)|$, hence Γ is a twisted cubic. □

The next proposition follows from a result of Cassels, see [Cor76a, Proposition 3.1, pp. 274–275]. We give a complete argument for convenience of the reader.

Proposition 2.4 (Cassels)**.** *A smooth cubic surface S/k has an effective zero-cycle of degree four if and only if it has an effective zero-cycle of degree five. More precisely, a closed point of either degree yields either a rational point or a closed point of the other degree.*

Proof. If $S(k) \neq \varnothing$, multiples of a rational point give both cycles. Otherwise Lemma 2.1 excludes points of degrees one and two; hence an effective zero-cycle of degree $d \in \{4, 5\}$ must be a single closed point P. By Lemma 2.2, its geometric points are in general linear position.

We claim that P lies on a twisted cubic over k. To see this, write $E = \kappa(P) = k(\theta)$ for a primitive element θ and let

$$\nu_3 : \mathbb{P}^1_k \longrightarrow \mathbb{P}^3_k, \qquad [s : t] \longmapsto [s^3 : s^2t : st^2 : t^3].$$

For $d = 4$, choose homogeneous coordinates $P = [a_0 : a_1 : a_2 : a_3]$ with $a_i \in E$. Noncoplanarity means that the a_i form a k-basis of E. Comparing this basis with $1, \theta, \theta^2, \theta^3$ gives a matrix in $\mathrm{GL}_4(k)$ taking $[1 : \theta : \theta^2 : \theta^3]$ to P. Its image of $\nu_3(\mathbb{P}^1)$ is the required curve.

For $d = 5$, let $\theta_1, \ldots, \theta_5$ be the conjugates of θ in a Galois splitting field. The five points $[1 : \theta_i : \theta_i^2 : \theta_i^3]$ for $i = 1, \ldots, 5$ form a projective frame, by the Vandermonde determinant.

The corresponding conjugates of P also form a projective frame. There is a unique projectivity matching these two labelled frames. Both labellings have the same Galois action, so uniqueness makes the projectivity Galois invariant; it therefore descends to k. Again its image of $\nu_3(\mathbb{P}^1)$ is a twisted cubic Γ containing P.

If $\Gamma \subset S$, the curve supplies a k-point. Otherwise Bézout's theorem gives

$$\deg(S \cap \Gamma) = 9, \qquad [S \cap \Gamma] - [P] \text{ effective of degree } 9 - d. \tag{2.1}$$

Under the standing assumption $S(k) = \varnothing$, this effective cycle is a single closed point, as above. □

3. The degree-five argument: Pfaffian presentations

In Section 3.1 we work over a field of characteristic different from two; then in Section 3.2 we restrict ourselves to characteristic zero. Our goal is to prove Theorem 1.4 stated in the introduction and show that it implies the following theorem.

Theorem 3.1. *A smooth cubic surface over a field of characteristic zero with an effective zero-cycle of degree five has a rational point.*

3.1. **Pfaffian cubics have rational points.** Let $V = k^6$ and put $\mathcal{T} = (\bigwedge^2 V^*)^3 \simeq \mathbb{A}^{45}_k$. For an alternating form $A \in \bigwedge^2 V^*$, a subspace $W \subset V$ is isotropic if $A|_{W\times W} = 0$.

Consider the scheme

$$\mathcal{I} = \{(W, A_0, A_1, A_2) \;\mid\; A_i|_{W\times W} = 0 \ (0 \le i \le 2)\} \subset \mathrm{Gr}(3, V) \times \mathcal{T}$$

which parametrizes three-spaces $W \subset V$ that are isotropic for each entry of $(A_0, A_1, A_2) \in \mathcal{T}$. The following lemma is an ordered-triple version of Han's incidence calculation [Han14, Proposition 2.4] (who credits Sasha Kuznetsov).

Lemma 3.2. *There is a nonempty open subset $\mathcal{T}^\circ \subset \mathcal{T}$, defined over k, such that the natural map $\mathcal{I} \times_{\mathcal{T}} \mathcal{T}^\circ \to \mathcal{T}^\circ$ is finite étale of degree two.*

Proof. For a three-space $W \subset V$, the restriction map $\bigwedge^2 V^* \to \bigwedge^2 W^*$ is surjective with 12-dimensional kernel. Thus $\mathcal{I}$ is the total space of a rank-36 vector bundle over $\mathrm{Gr}(3, V) = \mathrm{Gr}(3, 6)$. In particular, $\mathcal{I}$ is smooth of dimension 45 and the projection $\pi : \mathcal{I} \to \mathcal{T}$ is proper.

Write $V = V_1 \oplus V_2 \oplus V_3$, with $\dim V_i = 2$, and set

$$I := \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad J := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \qquad \text{and} \qquad D := \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Consider the triple of alternating 6×6 matrices

$$B_0 := \mathrm{diag}(J, J, J), \qquad B_1 := \mathrm{diag}(J, 2J, 3J), \qquad B_2 := \begin{pmatrix} 0 & I & I \\ -I & 0 & D \\ -I & -D & 0 \end{pmatrix}. \tag{3.1}$$

We aim to compute the fiber of π at (B_0, B_1, B_2) scheme-theoretically. A B_0-isotropic three-space W is Lagrangian with respect to the standard symplectic form on $V = k^6$. Put $T = B_0^{-1}B_1 = \mathrm{diag}(I, 2I, 3I)$. If W is also B_1-isotropic, then

$$B_0(Tw, w') = B_1(w, w') = 0 \qquad \text{for all } w, w' \in W.$$

Hence, $T(W) \subset W^{\perp_{B_0}} = W$. Therefore, W is T-invariant and splits into its intersections with the three T-eigenspaces. Isotropy bounds the dimension of each summand by one, so $W = \ell_1 \oplus \ell_2 \oplus \ell_3$ with $\ell_i \subset V_i$ a line. This description holds scheme-theoretically. The common isotropic locus for B_0, B_1 is therefore $(\mathbb{P}^1)^3$. Writing $\ell_i = [a_i : b_i]$ and imposing isotropy with respect to B_2, we obtain the following description of the fiber under consideration:

$$a_1a_2 + b_1b_2 = 0, \qquad a_1a_3 + b_1b_3 = 0, \qquad a_2a_3 + 2b_2b_3 = 0.$$

No a_i vanishes at a geometric solution: if $a_1 = 0$ then $b_2 = b_3 = 0$, contradicting the third equation; the other cases are similar. Set $u_i = b_i/a_i$. The above equations thus simplify to

$$u_2 = u_3 = -u_1^{-1}, \qquad u_1^2 + 2 = 0. \tag{3.2}$$

Hence the scheme-theoretic fiber of π at (B_0, B_1, B_2) is finite and geometrically reduced of degree two.

At these two geometric points the differential of π is an isomorphism, since source and target are smooth of the same dimension and the fiber tangent spaces vanish. Thus π is étale there. Properness allows us to remove the image of its non-étale locus and shrink around (B_0, B_1, B_2); the resulting morphism is proper and quasi-finite, hence finite étale, of degree two. □

Proof of Theorem 1.4. Let k be a field of characteristic different from 2 and let $F(x) = \mathrm{Pf}(A)$ be the Pfaffian of an alternating 6×6 matrix of linear forms in $x_0, \dots, x_3$ over k. We may write

$$F(x) = \mathrm{Pf}\left(\sum_{i=0}^{3} x_i A_i\right),$$

where the A_i are alternating 6×6 matrices over k. The theorem is trivial if F is identical to zero. We may therefore assume that $S = \{F = 0\} \subset \mathbb{P}^3_k$ is a cubic surface.

We first deal with the case where $(A_0, A_1, A_2) \in \mathcal{T}^\circ(k)$. Then Lemma 3.2 gives an extension L/k of degree at most two and a three-space $W \subset L^6$ isotropic for A_0, A_1, A_2.

The alternating form $A_3|_W$ has a nonzero radical vector $v \in W$, since $\dim W = 3$. Consequently all four linear forms

$$A_0 v,\ A_1 v,\ A_2 v,\ A_3 v$$

belong to the three-dimensional subspace $\mathrm{Ann}(W) \subset (L^6)^*$ of linear forms $\lambda\colon L^6 \to L$ that vanish on W. Choose a nontrivial relation $\sum c_i (A_i v) = 0$ with $c_i \in L$. Since $(\sum c_i A_i)v = 0$, the alternating matrix $\sum c_i A_i$ is degenerate, so

$$F(c_0, c_1, c_2, c_3)^2 = \det\left(\sum_{i=0}^{3} c_i A_i\right) = 0.$$

This gives an L-point of $F = 0$. Its image in S has degree at most two, and Lemma 2.1 (whose proof in fact applies over any field of characteristic different from 2) gives a k-point.

For an arbitrary triple $(A_0, A_1, A_2) \in \mathcal{T}(k)$, use the fixed matrices (3.1) and put

$$A_i(t) = (1-t)A_i + tB_i \quad (0 \le i \le 2), \qquad A_3(t) = A_3, \qquad F_t(x) = \mathrm{Pf}\left(\sum_{i=0}^{3} x_i A_i(t)\right).$$

The generic triple belongs to $\mathcal{T}^\circ$ because its value at $t = 1$ does; moreover $F_t \neq 0$ over $k(t)$ since $F_0 = F \neq 0$. The preceding argument over $k(t)$ gives a $k(t)$-point of $\{F_t = 0\}$. Scale its homogeneous coordinates into the discrete valuation ring $k[t]_{(t)}$, with at least one a unit. Reduction modulo t then gives a k-point of $\{F = 0\}$. □

3.2. **Five points and the characteristic-zero theorem.**

Lemma 3.3 (Beauville)**.** *Let $Z \subset \mathbb{P}^3_k$ be a geometrically reduced subscheme of degree five whose geometric points are in general linear position. Then every cubic form $F \in k[x_0, \dots, x_3]$ vanishing along Z is the Pfaffian of an alternating 6×6 matrix $A = (a_{ij})$ of linear forms $a_{ij} \in k[x_0, \dots, x_3]$.*

Proof. For a smooth cubic surface (which is the only case needed in this section), this is [Beau00, Proposition 7.2 and Examples 7.4]. The alternative approach suggested in [Beau00, Remark 7.3(b)] works for any cubic form. For clarity, we provide the proof of this alternative version below. See also Tanturri [Tan14, Section 2], who gives an explicit algorithm for this Pfaffian presentation starting from five k-rational points in general linear position.

Put $S = k[x_0, x_1, x_2, x_3]$, and let $I_Z \subset S$ be the homogeneous ideal of Z. We claim that the quotient ring S/I_Z is Gorenstein, with graded free resolution

$$0 \longrightarrow S(-5) \xrightarrow{q^t} S(-3)^5 \xrightarrow{T} S(-2)^5 \xrightarrow{q} S \longrightarrow S/I_Z \longrightarrow 0, \tag{3.3}$$

where T is a 5×5 alternating matrix of linear forms,

$$q_i = (-1)^{i+1} \operatorname{Pf}(T_{\widehat{i}}) \qquad (1 \le i \le 5)$$

are the signed Pfaffians of the principal submatrices $T_{\widehat{i}}$ of T, and $q = (q_1, q_2, q_3, q_4, q_5)$.

Let us first prove these statements over an algebraic closure $\overline{k}$. The five geometric points can be moved to $[e_0], [e_1], [e_2], [e_3], [1:1:1:1]$. It is easy to see that the ideal $I_Z \otimes_k \overline{k} \subset S \otimes_k \overline{k}$ is generated by the following five quadratic polynomials:

$$\begin{aligned} q_1' &= x_0x_2 - x_0x_1, & q_2' &= x_0x_3 - x_0x_1, \\ q_3' &= x_1x_2 - x_0x_1, & q_4' &= x_1x_3 - x_0x_1, \\ q_5' &= x_2x_3 - x_0x_1. \end{aligned}$$

These are the signed principal Pfaffians of the alternating matrix

$$T' = \begin{pmatrix} 0 & -x_1 & -x_3 & 0 & x_1 \\ x_1 & 0 & 0 & x_2 & -x_1 \\ x_3 & 0 & 0 & x_0 & -x_0 \\ 0 & -x_2 & -x_0 & 0 & x_0 \\ -x_1 & x_1 & x_0 & -x_0 & 0 \end{pmatrix}$$

Put $\bar{S} = S \otimes_k \overline{k}$ and $q' = (q_1', \dots, q_5')$. Since the signed principal Pfaffians of T' generate the height-three ideal $I_Z\bar{S}$, the Buchsbaum–Eisenbud structure theorem for codimension 3 Gorenstein ideals [BE77, Theorem 2.1(1)] shows that $\bar{S}/I_Z\bar{S}$ is Gorenstein and that the associated Pfaffian complex (3.3) (with q', T' in place of q, T) is a free resolution.

Graded Betti numbers and Gorensteinness descend under field extensions. Thus S/I_Z is Gorenstein and its minimal graded free resolution has the same ranks and degree shifts.

By the graded version of [BE77, Theorem 2.1(2) and its proof] (cf. the discussion about the graded version on [BE77, page 466]), this resolution can be chosen to be a Pfaffian resolution as in (3.3). Its middle map $S(-3)^5 \to S(-2)^5$ is therefore represented by a 5×5 alternating matrix T of linear forms and the quadrics $q_i = (-1)^{i+1} \operatorname{Pf}(T_{\widehat{i}})$ and $q = (q_1, \dots, q_5)$ make (3.3) exact (over k).

In particular, I_Z is generated by $q_1, \dots q_5$. If F is a cubic vanishing on Z, we may thus write $F = \sum_{i=1}^5 \ell_i q_i$ with linear forms ℓ_i over k. Expanding along the last column gives

$$F = \operatorname{Pf} \begin{pmatrix} T & \ell^t \\ -\ell & 0 \end{pmatrix}, \qquad \ell = (\ell_1, \dots, \ell_5), \tag{3.4}$$

as we want. □

Proof of Theorem 3.1. If the cycle has a point of degree one or two in its support, apply Lemma 2.1. Otherwise it is a single closed point P of degree five. The coplanar case is Lemma 2.2(1). In the remaining case Lemma 2.2(2) gives general linear position, Lemma 3.3 supplies a Pfaffian presentation over k, and Theorem 1.4 gives a rational point. □

4. The degree-four argument: nets of quadrics

Combining Theorem 3.1 and Proposition 2.4 proves Theorem 1.1 stated in the introduction, which assumes $\operatorname{char}(k) = 0$. In this section we give a second proof, which naturally works over any perfect field and yields the following theorem.

Theorem 4.1. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over a perfect field. If S has a closed point P of degree four, then $S(k) \neq \varnothing$.*

4.1. **A rational-point criterion from three quadrics.** Let $S = \{F = 0\} \subset \mathbb{P}^3_k$ be a smooth cubic surface over a field k. If S has a k-rational point P, then we may after a change of coordinates assume that $P = [1 : 0 : 0 : 0]$, in which case $F = x_1Q_1 + x_2Q_2 + x_3Q_3$ for quadratic forms Q_i. Conversely, if

$$F = H_1Q_1 + H_2Q_2 + H_3Q_3 \in k[x_0, x_1, x_2, x_3] \tag{4.1}$$

for linear and quadratic forms $H_i \in k[x_0, x_1, x_2, x_3]$ and $Q_i \in k[x_0, x_1, x_2, x_3]$, respectively, then S contains the linear space $\{H_1 = H_2 = H_3 = 0\} \subset \mathbb{P}^3_k$ and hence a rational point.

Our strategy is to find a decomposition as in (4.1).

Lemma 4.2. *Let $S = \{F = 0\} \subset \mathbb{P}^3_k$ be a smooth cubic surface over a field k. Let $Z \subset S$ be a zero-dimensional subscheme of length 8 and assume that there are three quadratic forms $Q_1, Q_2, Q_3 \in k[x_0, x_1, x_2, x_3]$ such that $W := \{Q_1 = Q_2 = Q_3 = 0\} \subset \mathbb{P}^3_k$ has dimension zero and contains Z. Then there are linear forms $H_i \in k[x_0, x_1, x_2, x_3]$ such that (4.1) holds.*

Proof. Since W has dimension zero and contains Z, we must have $W = Z$, because W has length $2^3 = 8$. Hence, $\mathcal{I}_Z = (Q_1, Q_2, Q_3)$ and Q_1, Q_2, Q_3 form a regular sequence. The Koszul complex therefore yields the exact sequence

$$0 \longrightarrow \mathcal{O}(-6) \longrightarrow \mathcal{O}(-4)^{\oplus 3} \longrightarrow \mathcal{O}(-2)^{\oplus 3} \longrightarrow \mathcal{I}_Z \longrightarrow 0 \tag{4.2}$$

on $\mathbb{P}^3$. Twisting with $\mathcal{O}(3)$, this yields the exact sequence

$$0 \longrightarrow \mathcal{O}(-3) \longrightarrow \mathcal{O}(-1)^{\oplus 3} \longrightarrow \mathcal{O}(1)^{\oplus 3} \longrightarrow \mathcal{I}_Z(3) \longrightarrow 0.$$

Since $H^{i+1}(\mathbb{P}^3, \mathcal{O}(-3)) = H^i(\mathbb{P}^3, \mathcal{O}(-1)) = 0$ for $i = 0, 1$, we conclude that

$$H^0(\mathbb{P}^3, \mathcal{O}(1)^{\oplus 3}) \xrightarrow{\cong} H^0(\mathbb{P}^3, \mathcal{I}_Z(3)), \qquad (H_1, H_2, H_3) \mapsto \sum_{i=1}^{3} H_iQ_i$$

is an isomorphism. Finally, since $Z \subset S$, the cubic form F yields a global section of $\mathcal{I}_Z(3)$ and we obtain a decomposition as in (4.1). □

Remark 4.3. Twisting (4.2) with $\mathcal{O}(2)$ and taking cohomology, one obtains similarly that $H^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \cong H^0(\mathbb{P}^3, \mathcal{O}^{\oplus 3})$ has dimension 3 in the setting of Lemma 4.2.

Proposition 4.4. *Let $S = \{F = 0\}$ be a smooth cubic surface over a field k. Let $Z \subset S$ be a zero-dimensional subscheme of length 8. Assume that the base locus of the linear series $|\mathcal{I}_Z(2)|$ on $\mathbb{P}^3$ is zero-dimensional. Then S admits a rational point.*

Proof. Since the base locus of $|\mathcal{I}_Z(2)|$ is zero-dimensional, we can find, after base change to an infinite field, three linearly independent quadrics which have zero-dimensional intersection. Hence, $h^0(\mathbb{P}^3_k, \mathcal{I}_Z(2)) = h^0(\mathbb{P}^3_{\bar{k}}, \mathcal{I}_{Z_{\bar{k}}}(2)) = 3$ by Remark 4.3. It follows that there are three quadratic forms $Q_1, Q_2, Q_3 \in k[x_0, x_1, x_2, x_3]$ which satisfy the assumptions in Lemma 4.2. Hence, F has a decomposition as in (4.1) and so S has a rational point. □

A necessary condition for the base locus of $|\mathcal{I}_Z(2)|$ to be zero-dimensional is $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \geq 3$. We will study this condition and the resulting base loci in the next subsection; since base loci are compatible with the extension of the ground field, it will be enough to treat the geometric case where the ground field is algebraically closed.

4.2. **Quadrics through four tangent double points: a geometric analysis.** In this subsection we work over an algebraically closed field k. Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface and let $p_0, \dots, p_3 \in S$ be closed points that do not lie on a plane. Up to a coordinate change, we may assume that $p_i = [e_i]$ are the four coordinate vertices in $\mathbb{P}^3_k$, i.e. the points represented by the standard basis in k^4. Let $Z = \sqcup_{i=0}^3 Z_i$ be a length 8 subscheme of S, where Z_i is of length 2 and supported at the point p_i.

We may represent Z_i by $[e_i + \epsilon v_i]$, where $\epsilon^2 = 0$ and $v_i = (u_{i0}, \dots, u_{i3})$ is a nontrivial tangent direction at p_i with $u_{ii} = 0$. That is, the line $\{[se_i + tv_i] \mid [s : t] \in \mathbb{P}^1\} \subset \mathbb{P}^3$

is contained in the projective tangent space of S at p_i. In the affine chart $x_i = 1$, with coordinates $y_j = x_j/x_i$ for $j \neq i$, the tangent direction v_i may equivalently be described by

$$v_i = \sum_{j\neq i} u_{ij} \frac{\partial}{\partial y_j} \in T_{p_i} S. \tag{4.3}$$

Lemma 4.5. *In the above notation, we have $h^0(\mathbb{P}^3_k, \mathcal{I}_Z(2)) \geq 3$ if and only if there are scalars $\lambda_0, \ldots, \lambda_3 \in k$, not all zero, such that*

$$\lambda_i u_{ij} = -\lambda_j u_{ji} \qquad \text{for all } 0 \leq i < j \leq 3.$$

Proof. A quadric through $p_0, \ldots, p_3$ has the form

$$Q = \sum_{0\leq i<j\leq 3} q_{ij} x_i x_j. \tag{4.4}$$

The space of such quadrics has dimension $\binom{4}{2} = 6$.

For convenience, set $q_{ji} = q_{ij}$ whenever $i < j$. The quadric Q in (4.4) vanishes along Z_i if and only if $d_{p_i} Q(v_i) = 0$. Moreover,

$$d_{p_i} Q(v_i) = \sum_{j\neq i} q_{ij} u_{ij}. \tag{4.5}$$

Thus the quadrics vanishing along Z form the kernel of the linear map

$$H^0(\mathbb{P}^3, \mathcal{I}_{Z_{\mathrm{red}}}(2)) \longrightarrow k^4, \qquad Q \mapsto (d_{p_0} Q(v_0), \ldots, d_{p_3} Q(v_3))$$

from a six-dimensional vector space to k^4. This kernel has dimension at least 3 if and only if the four linear functionals $Q \mapsto d_{p_i} Q(v_i)$ are linearly dependent. This is equivalent to the existence of scalars λ_i, not all zero, such that

$$0 = \sum_{i=0}^{3} \lambda_i d_{p_i} Q(v_i) = \sum_{i<j} q_{ij} (\lambda_i u_{ij} + \lambda_j u_{ji}) \tag{4.6}$$

for every Q. Since the coefficients q_{ij} are arbitrary, this is equivalent to $\lambda_i u_{ij} = -\lambda_j u_{ji}$ for every $i < j$, as we want. □

The previous lemma gave a necessary and sufficient condition for $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \geq 3$. We will concentrate on the case $u_{ij} = -u_{ji}$ for all i, j, which ensures $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \geq 3$. We further assume $u_{ij} \neq 0$ for $i \neq j$ to rule out degenerate cases (which will be dealt with later in Proposition 4.8 via a different argument). The next lemma characterizes the possible base loci in this situation.

Lemma 4.6. *Assume in the above notation that $u_{ij} = -u_{ji}$ for all i, j and $u_{ij} \neq 0$ for $i \neq j$. Then the base locus B of the linear series $|\mathcal{I}_Z(2)| \subset |\mathcal{O}_{\mathbb{P}^3}(2)|$ is one of the following:*

(1) $B = Z$, or
(2) B is a twisted cubic in $\mathbb{P}^3$.

Proof. Consider the matrix $U := (u_{ij}) \in k^{4\times 4}$ which encodes the tangent directions of Z at the four coordinate vertices. For convenience, we rename the u_{ij} by a, b, c, d, e, f and get

$$U = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix}, \qquad \text{with} \quad abcdef \neq 0.$$

A direct computation using (4.5) shows that the quadratic forms

$$Q_1 \coloneqq \frac{d}{a}x_0x_1 - \frac{d}{b}x_0x_2 + x_1x_2,$$
$$Q_2 \coloneqq \frac{e}{a}x_0x_1 - \frac{e}{c}x_0x_3 + x_1x_3,$$
$$Q_3 \coloneqq \frac{f}{b}x_0x_2 - \frac{f}{c}x_0x_3 + x_2x_3,$$

are linearly independent and satisfy $d_{p_i}Q_j(v_i) = 0$ for all $i = 0, 1, 2, 3$ and all $j = 1, 2, 3$. Hence, $Q_1, Q_2, Q_3 \in H^0(\mathbb{P}^3, \mathcal{I}_Z(2))$ span a three-dimensional subspace.

We aim to describe the base locus $B' \coloneqq \{Q_1 = Q_2 = Q_3 = 0\}$. If this is zero-dimensional, then B must be equal to Z. It thus suffices to show that B' is either zero-dimensional or a twisted cubic. Indeed, if B' is a twisted cubic, then so is B, because zero-dimensional B implies $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) = 3$ by Remark 4.3 (cf. proof of Proposition 4.4).

On the plane $x_0 = 0$, the common zero-set of these quadrics is given by the coordinate vertices p_1, p_2, p_3. It remains to analyze the affine chart $x_0 = 1$ and we write $x_1 = t$. The vanishing $Q_1 = Q_2 = 0$ gives

$$x_2 = \frac{bdt}{a(d - bt)} \qquad \text{and} \qquad x_3 = \frac{cet}{a(e - ct)}. \tag{4.7}$$

One checks that neither denominator can vanish at a point of the base locus B'. Substitution into the third equation gives

$$Q_3 = \frac{t^2\Delta}{a^2(d - bt)(e - ct)} \qquad \text{where} \quad \Delta \coloneqq abef - acdf + bcde.$$

If $\Delta \neq 0$, the base scheme B' is therefore finite. In fact, the above equation gives a base scheme of length 2 on the chart $x_0 = 1$, which is compatible with the fact that we work away from p_1, p_2, p_3.

Suppose now that $\Delta = 0$. Then the vanishing of Q_3 is automatic, once (4.7) holds. Moreover, $be - cd \neq 0$, since otherwise $\Delta = bcde \neq 0$. We introduce a second variable s, homogenize (4.7) with respect to s, and clear denominators to find that the base locus is contained in the projective curve described by

$$x_0 = as(ds - bt)(es - ct),$$
$$x_1 = at(ds - bt)(es - ct),$$
$$x_2 = bdst(es - ct),$$
$$x_3 = cest(ds - bt),$$

with $[s : t] \in \mathbb{P}^1$. Note that each x_i is homogeneous of degree 3 in s, t, hence may be viewed as an element of $H^0(\mathbb{P}^1, \mathcal{O}(3))$. We check that the coefficient matrix of $x_0, \dots, x_3$ in the basis s^3, s^2t, st^2, t^3 has determinant $-a^2b^2c^2d^2e^2(be - cd) \neq 0$. Hence, the above expressions for $x_0, \dots, x_3$ form a basis of $H^0(\mathbb{P}^1, \mathcal{O}(3))$, hence they parametrize a smooth twisted cubic $\Gamma \subset \mathbb{P}^3$.

To double check our computation, we substitute the above parametrization of Γ into Q_1 and Q_2 and get zero, while substitution into Q_3 gives

$$s^2t^2(ds - bt)(es - ct)\Delta = 0.$$

Thus all three quadrics vanish on Γ. The ideal sheaf of a twisted cubic is always generated by three quadrics (see [Eis05, Proposition 6.1]) and so we must have $\mathcal{I}_\Gamma = (Q_1, Q_2, Q_3)$. This concludes the proof of the lemma. □

4.3. **First arithmetic consequences.** We first show that the assumption $u_{ij} = -u_{ji}$ used in Lemma 4.6 is automatic in the arithmetic situation, as follows.

Lemma 4.7. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over a perfect field k, and let $Z \subset S$ be a subscheme of length 8 supported at a closed point P of degree 4 whose geometric points span $\mathbb{P}^3_{\bar{k}}$. If $h^0(\mathbb{P}^3_k, \mathcal{I}_Z(2)) \geq 3$, then, over $\bar{k}$, and in coordinates such that Z is supported on the coordinate vertices $p_i = [e_i]$, the tangent directions can be chosen so that $u_{ij} = -u_{ji}$ for all i, j in (4.3).*

Proof. We work over $\bar{k}$ and pick coordinates so that the geometric points of the support of Z are the four coordinate vertices $p_i = [e_i]$. Lemma 4.5 shows that

$$R := \Big\{(\lambda_i) \in \bar{k}^4 \;\mid\; \lambda_i u_{ij} + \lambda_j u_{ji} = 0 \text{ for all } i < j\Big\}$$

is nonzero.

Write $f_i(Q) = d_{p_i}Q(v_i)$ for the functionals (4.5) in Lemma 4.5. By (4.6), R is the space of relations $\sum_i \lambda_i f_i = 0$. Since Z is defined over k, the natural Galois action on $H^0(\mathbb{P}^3_{\bar{k}}, \mathcal{I}_{Z_{\mathrm{red}}}(2))^\vee$ satisfies, for all $\sigma \in \mathrm{Gal}(\bar{k}/k)$,

$$\sigma(f_i) = c_{\sigma,i} f_{\sigma(i)}$$

for some $c_{\sigma,i} \in \bar{k}^\times$, where $\sigma(p_i) = p_{\sigma(i)}$. Applying σ to a relation $\lambda \in R$ therefore induces a semilinear action on R, given explicitly by

$$(\sigma \cdot \lambda)_{\sigma(i)} = c_{\sigma,i}\, \sigma(\lambda_i).$$

In particular, this action permutes the coordinate hyperplanes according to the transitive Galois action on the four points. Thus, if R were contained in one coordinate hyperplane $\{\lambda_i = 0\}$, it would be contained in all four, contradicting $R \neq 0$.

Since $\bar{k}$ is infinite, the four proper linear subspaces $R \cap \{\lambda_i = 0\}$ $(i = 0, 1, 2, 3)$ cannot cover R. We may therefore choose $(\lambda_i) \in R$ with $\lambda_i \neq 0$ for all i. Replacing v_i by $\lambda_i v_i = (\widetilde{u}_{i0}, \ldots, \widetilde{u}_{i3})$ preserves Z_i and gives

$$\widetilde{u}_{ij} = \lambda_i u_{ij} = -\lambda_j u_{ji} = -\widetilde{u}_{ji},$$

as required. □

Lemma 4.6 describes the base locus in the case where $u_{ij} = -u_{ji}$ is nonzero for $i \neq j$. The missing case will be handled through the following ad hoc argument.

Proposition 4.8. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over a perfect field k, and let $P \in S$ be a closed point of degree 4 whose geometric points span $\mathbb{P}^3_{\bar{k}}$. Let $Z \subset S$ be a subscheme of length 8 with $Z_{\mathrm{red}} = P$. Suppose that, after choosing coordinates in which the four geometric points of P are the coordinate vertices $p_i = [e_i]$, we can write*

$$Z_{\bar{k}} = \coprod_{i=0}^{3} Z_i, \qquad Z_i = \{[e_i + \epsilon v_i] \;\mid\; \epsilon^2 = 0\}, \qquad v_i = \sum_{j=0}^{3} u_{ij} e_j,$$

where every v_i is nonzero and $U = (u_{ij})$ is skew-symmetric with $u_{ii} = 0$ for all i.

If $u_{ij} = 0$ for some $i \neq j$, then $S(k) \neq \varnothing$.

Proof. Consider the graph G on the set of vertices $\{0, 1, 2, 3\}$ whose edges are the unordered pairs $\{i, j\}$ with $u_{ij} \neq 0$. This graph carries a natural action by the absolute Galois group of k. The Galois action on the four vertices is transitive, so this graph is regular, i.e. there is a positive integer n such that its valence at each vertex is n. Since every v_i is nonzero, while some u_{ij} with $i \neq j$ are zero, we find $n \in \{1, 2\}$.

Suppose first that $n = 1$. After relabeling, the two edges are $\{0, 1\}$ and $\{2, 3\}$. For $i \neq j$, write $L_{ij} = \langle p_i, p_j \rangle$ for the line spanned by p_i and p_j. Then the subscheme $Z_0 \sqcup Z_1$ is contained in L_{01}, so the restriction of the cubic equation of S to L_{01} vanishes on a subscheme of length 4. Hence, $L_{01} \subset S_{\bar{k}}$. A similar argument shows $L_{23} \subset S_{\bar{k}}$.

The lines L_{01} and L_{23} have no point in common and their union is Galois invariant. In fact, the Galois action interchanges the two lines and so their union is defined over k and there is a quadratic extension K/k over which L_{01} and L_{23} are defined. Cubic surfaces that contain such skew lines over a quadratic extension are rational, hence contain in particular

a rational point. To construct such a point explicitly, choose $a \in L_{01}(K)$ and a generator $\sigma \in \mathrm{Gal}(K/k)$. Then the line $M = \langle a, \sigma(a)\rangle$ is defined over k. If $M \subset S$, it supplies a k-rational point. Otherwise the residual intersection point of $M \cap S$ provides such a point.

Suppose now that $n = 2$. Then G is the four-cycle graph C_4. After relabeling, we may assume that

$$u_{01} = u_{23} = 0, \qquad b = u_{02}, \quad c = u_{03}, \quad d = u_{12}, \quad e = u_{13}, \qquad bcde \neq 0.$$

The three quadrics

$$Q_1 = x_0x_1, \qquad Q_2 = x_2x_3, \qquad Q_3 = \frac{x_0x_2}{b} - \frac{x_0x_3}{c} - \frac{x_1x_2}{d} + \frac{x_1x_3}{e}$$

vanish on $Z_{\bar{k}}$. Indeed, they vanish at the four vertices, and direct substitution using (4.5) gives $d_{p_i}Q_j(v_i) = 0$ for every i, j. Moreover,

$$\{Q_1 = Q_2 = 0\} = L_{02} \cup L_{03} \cup L_{12} \cup L_{13}.$$

The form Q_3 restricts nontrivially on each of these lines. Thus $\{Q_1 = Q_2 = Q_3 = 0\}$ is zero-dimensional. In particular, the base locus of $|\mathcal{I}_{Z_{\bar{k}}}(2)|$ is zero-dimensional.

Formation of the base locus commutes with extension of the ground field, so the base locus of $|\mathcal{I}_Z(2)|$ over k is also zero-dimensional and Proposition 4.4 gives $S(k) \neq \varnothing$. □

4.4. **Construction of the length-8 subscheme.** We now construct a subscheme $Z \subset S$ of length 8 over the ground field by choosing a suitable tangent direction at a closed point of degree 4.

Proposition 4.9. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over a perfect field k, and let $P \in S$ be a closed point of degree 4 whose geometric points span $\mathbb{P}^3_{\bar{k}}$. Then there exists a subscheme $Z \subset S$, defined over k and supported at P, such that $Z_{\bar{k}}$ is a disjoint union of four subschemes of length 2 and*

$$h^0(\mathbb{P}^3_k, \mathcal{I}_Z(2)) \geq 3.$$

Proof. Put $L = \kappa(P)$, and let $\mathcal{J}_P \subset \mathcal{O}_S$ be the ideal sheaf of P in S. Since S is smooth and L/k is separable, the restriction of the cotangent bundle $\Omega^1_S \coloneqq \Omega^1_{S/k}$ to P,

$$\Omega^1_S|_P \coloneqq \Omega^1_{S/k} \otimes_{\mathcal{O}_S} L \cong \mathcal{J}_P/\mathcal{J}_P^2,$$

has dimension 2 over L. The dual of this space is the tangent space $T_S|_P$ at P.

The choice of a subscheme $Z \subset S$ of length 8 which is supported at the point P and whose base change to $\bar{k}$ splits into 4 subschemes of length 2 is equivalent to the choice of a line in the L-vector space $T_S|_P$. Dually, it corresponds to a one-dimensional quotient of $\Omega^1_S|_P$.

Set

$$E \coloneqq \Omega^1_S(2)|_P \qquad \text{and} \qquad V \coloneqq H^0(\mathbb{P}^3_k, \mathcal{I}_P(2)),$$

where $\mathcal{I}_P$ is the ideal sheaf of P in $\mathbb{P}^3_k$. The four geometric points of P impose independent conditions on quadrics, since they span $\mathbb{P}^3_{\bar{k}}$. Thus $\dim_k V = 6$. Moreover, $\dim_L E = 2$ and $\dim_k E = 8$. Restricting a quadric which vanishes at P to S and reducing modulo $\mathcal{J}_P^2$ defines a canonical k-linear map

$$\varphi\colon V \longrightarrow H^0(P, (\mathcal{J}_P/\mathcal{J}_P^2) \otimes_L \mathcal{O}_S(2)|_P) \cong \Omega^1_S(2)|_P,$$

where we use that $\Omega^1_S(2)|_P \cong (\mathcal{J}_P/\mathcal{J}_P^2) \otimes_L \mathcal{O}_S(2)|_P$.

Let us first express the desired property in terms of this map. Given any L-linear quotient $q\colon E \twoheadrightarrow N$ with $\dim_L N = 1$, tensoring with $\mathcal{O}_S(-2)|_P$ yields a one-dimensional quotient of $\Omega^1_S|_P$, hence a length 8 subscheme Z_q supported at P as above. A quadric $Q \in V$ vanishes on Z_q precisely when $q(\varphi(Q)) = 0$. Therefore

$$H^0(\mathbb{P}^3_k, \mathcal{I}_{Z_q}(2)) = \ker\bigl(q \circ \varphi\colon V \longrightarrow N\bigr). \tag{4.8}$$

Since $\dim_k V = 6$ and $\dim_k N = 4$, the desired inequality $h^0(\mathbb{P}^3_k, \mathcal{I}_{Z_q}(2)) \geq 3$ is equivalent to $\mathrm{rank}_k(q \circ \varphi) \leq 3$. Equivalently, there must exist a nonzero k-linear functional $\mu\colon N \to k$ such

that $\mu \circ q \circ \varphi = 0$. Pulling this functional back along q gives a nonzero k-linear functional $\lambda := \mu \circ q \colon E \to k$ satisfying

$$\lambda(\varphi(Q)) = 0 \qquad \text{for every } Q \in V. \tag{4.9}$$

We will first choose such a functional λ and then recover an L-linear one-dimensional quotient of E through which it factors.

Consider the k-vector space

$$K = \ker\left(\varphi^\vee \colon \operatorname{Hom}_k(E,k) \longrightarrow \operatorname{Hom}_k(V,k)\right), \qquad \varphi^\vee(\lambda) = \lambda \circ \varphi.$$

The source has dimension 8 and the target dimension 6, so $\dim_k K \geq 2$. We may therefore choose a nontrivial element $\lambda \in K$. This gives a k-linear functional on E which satisfies (4.9).

It remains to show that the k-linear map λ factors through some one-dimensional L-linear quotient $q \colon E \to N \cong L$ of E. To this end, we use the k-linear isomorphism

$$\begin{aligned} \Theta : \operatorname{Hom}_L(E,L) &\xrightarrow{\sim} \operatorname{Hom}_k(E,k), \\ f &\mapsto \operatorname{Tr}_{L/k} \circ f, \end{aligned} \tag{4.10}$$

provided by the trace map. (To see that Θ is an isomorphism of k-vector spaces, it suffices to note that it is k-linear and injective, while source and target have the same finite dimension over k. To see injectivity, note that any nonzero L-linear map $f \colon E \to L$ is surjective and $\operatorname{Tr}_{L/k}$ is surjective because L/k is separable, hence $\operatorname{Tr}_{L/k} \circ f$ is surjective and therefore nonzero as long as f is nonzero.)

It follows that there is a unique nonzero L-linear map $f \colon E \to L$ such that $\lambda = \operatorname{Tr}_{L/k} \circ f$. Since λ is nonzero, f must be surjective, and λ factors through it; the role of $\mu \colon N \cong L \to k$ above is then taken by the trace map $\operatorname{Tr}_{L/k}$. As explained above, the length 8 subscheme that is associated to the one-dimensional quotient f has then the properties asserted in the proposition. This concludes the proof. □

4.5. **Proof of Theorem 4.1.**

Proof of Theorem 4.1. By Lemma 2.2, we may assume that the geometric points associated to P span $\mathbb{P}^3_{\bar{k}}$. By Proposition 4.9, there is a closed subscheme $Z \subset S$ of length 8 supported on P such that $Z_{\bar{k}}$ is a disjoint union of four subschemes of length 2 and such that $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \geq 3$.

We pass to the algebraic closure $\bar{k}$ of k and choose coordinates on $\mathbb{P}^3_{\bar{k}}$ such that the four geometric points $p_0, \dots, p_3$ that correspond to P agree with the four coordinate vertices: $p_i = [e_i]$. We may then write $Z_{\bar{k}} = \sqcup_{i=0}^3 Z_i$, where Z_i is the length 2 subscheme supported at p_i that is given by $[e_i + \epsilon v_i]$ with $\epsilon^2 = 0$ and $v_i = (u_{i0}, \dots, u_{i3})$ with $u_{ii} = 0$. By Lemma 4.7, we can arrange that $u_{ij} = -u_{ji}$. By Proposition 4.8, we may further assume that u_{ij} is nonzero for all $i \neq j$. It then follows from Lemma 4.6 that the base locus of $|\mathcal{I}_{Z_{\bar{k}}}(2)|$ is either finite or a twisted cubic. It follows that the base locus of the linear series $|\mathcal{I}_Z(2)|$ on $\mathbb{P}^3_k$ over the ground field k is either finite or a smooth curve Γ whose base change to $\bar{k}$ is a twisted cubic, hence $\Gamma \cong \mathbb{P}^1_k$ is a twisted cubic by Lemma 2.3.

If the base locus of $|\mathcal{I}_Z(2)|$ is zero-dimensional, then S has a rational point by Proposition 4.4. Otherwise, the base locus is a twisted cubic Γ. If $\Gamma \subset S$, then S has a rational point because $\Gamma \cong \mathbb{P}^1_k$. Otherwise, $\Gamma \cap S$ is a zero-dimensional subscheme of length 9 which contains the length 8 subscheme Z; the residual intersection point then yields a rational point. This concludes the proof of the theorem. □

5. From characteristic zero to positive characteristic

We begin with the following.

Theorem 5.1. *A smooth cubic surface over a field of characteristic zero with a zero-cycle of degree one has a rational point.*

Proof. Voisin's theorem [Voi26, Theorem 1.5], which improved Coray's reduction [Cor76a, Theorem 7.1], gives a point of degree one or four. We may then either apply Theorem 4.1, or use Proposition 2.4 to produce an effective zero-cycle of degree five and apply Theorem 3.1. □

In the remainder of this section, we reduce the case of arbitrary base fields (including imperfect ones) to the case of characteristic zero. The required lifting statement follows [Ma21, Section 2.2, Lemmas 2.4–2.5]; we include its proof.

Lemma 5.2. *Let k have characteristic $p > 0$, let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface, and let $P \subset S$ be a closed point of degree d. There exist a complete discrete valuation ring R of characteristic zero with residue field k, a smooth cubic surface $\mathscr{S} \subset \mathbb{P}^3_R$ with special fiber S, and a finite flat subscheme $\mathscr{Z} \subset \mathscr{S}$ of degree d with special fiber P.*

Proof. Choose a Cohen ring R with uniformizer p and residue field exactly k [Stacks, Tag 0328]. Lift the coefficients of a defining cubic of S to obtain $\mathscr{S} \subset \mathbb{P}^3_R$. This hypersurface is flat over R and smooth along its special fiber. Its nonsmooth locus is closed and proper over R. If nonempty, its image would contain the closed point of $\operatorname{Spec} R$, a contradiction. Thus $\mathscr{S}/R$ is smooth.

Since R is a discrete valuation ring and $\mathcal{S}/R$ is smooth and projective of relative dimension two, Fogarty's theorem [Fog68, Theorem 2.9] implies that $\operatorname{Hilb}^d(\mathcal{S}/R)$ is smooth over R, see also [Stacks, Tag 06D9] and [Ma21, Lemma 2.5]. As R is complete, hence henselian, the k-point $[P]$ lifts to an R-point. The corresponding universal family gives the required finite flat subscheme $\mathcal{Z} \subset \mathcal{S}$. □

We are now ready to prove the case of smooth cubic surfaces.

Theorem 5.3. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over an arbitrary field. If S has a zero-cycle of degree one, then $S(k) \neq \varnothing$.*

Proof. By Theorem 5.1, only positive characteristic remains. Choose a closed point P of degree d prime to three from the support of a zero-cycle of degree one. Apply Lemma 5.2, and put $K = \operatorname{Frac}(R)$. The finite scheme $\mathscr{Z}_K$ has degree d, so its fundamental cycle, together with the degree-three intersection class with a line, gives a zero-cycle of degree one on $\mathscr{S}_K$.

Theorem 5.1 gives a K-point of $\mathscr{S}_K$. Properness extends it to an R-point of $\mathscr{S}$, whose reduction is a k-point of S. □

6. Removing the smoothness hypothesis

In this section we finish the proof of the following theorem, which is equivalent to Theorem 1.2.

Theorem 6.1. *Let $X = \{F = 0\} \subset \mathbb{P}^3_k$ be a cubic surface over a field k. Then X has a k-rational point if and only if it has a closed point of degree coprime to 3, or, equivalently, a zero-cycle of degree one.*

For singular cubic surfaces over perfect fields, the conclusion of Theorem 6.1 is due to Coray [Cor76a, Proposition 8.2]; see also [Cor76b, Corollary 1] for a second proof. We give a reduction to the smooth case over a Laurent-series field that applies over arbitrary fields.

Let $X = \{F = 0\} \subset \mathbb{P}^3_k$ for a nonzero homogeneous cubic F, with no assumption on k or on the singularities of X. If F is reducible over k, it has a linear factor ℓ, so X contains the k-plane $\{\ell = 0\}$ and $X(k) \neq \varnothing$. Write X_{sm} for the smooth locus over k. We first handle the possibility that a zero-cycle of degree one is supported entirely in the singular locus.

Lemma 6.2. *Let L/k be a finite extension with $3 \nmid [L : k]$. If $X(L) \neq \varnothing$, then either $X(k) \neq \varnothing$ or $X_{\mathrm{sm}}(L) \neq \varnothing$.*

Proof. Choose $p \in X(L)$. Its multiplicity on the hypersurface X_L is one, two, or three. Multiplicity one means that p is smooth. If p has multiplicity two, we will construct a smooth L-rational point; if it has degree 3 we will show that X has a k-rational point.

Suppose the multiplicity is two. In coordinates over L with $p = [1:0:0:0]$, the equation is

$$F = x_0 Q(x_1, x_2, x_3) + C(x_1, x_2, x_3), \qquad Q \neq 0,$$

where Q and C have degrees two and three, respectively. There exists $a \in L^3$ with $Q(a) \neq 0$, even over a finite field, because the values $Q(e_i)$ and $Q(e_i + e_j) - Q(e_i) - Q(e_j)$ recover all coefficients of Q. Then the point

$$q = \left[-\frac{C(a)}{Q(a)} : a_1 : a_2 : a_3 \right]$$

lies on X_L and satisfies $\partial F/\partial x_0(q) = Q(a) \neq 0$. It is therefore a smooth point of X_L, and so $X_{\mathrm{sm}}(L) \neq \emptyset$.

Suppose now that p has multiplicity three. In coordinates with $p = [1:0:0:0]$, the equation F that cuts out X is then independent of x_0. In the original coordinates, a nonzero vector $v \in L^4$ representing p consequently satisfies

$$F(x + sv) = F(x) \quad \text{in } L[x_0, x_1, x_2, x_3, s]. \tag{6.1}$$

If $\operatorname{char} k \neq 3$, consider the k-linear map

$$D_F \colon k^4 \longrightarrow k[x_0, x_1, x_2, x_3]_2, \qquad w \longmapsto D_w F = \sum_{i=0}^{3} w_i \frac{\partial F}{\partial x_i}.$$

By differentiating (6.1) with respect to s, we obtain $D_v F = 0$. Thus $D_F \otimes_k L$ has nonzero kernel, and hence so does D_F. We may thus pick a nonzero vector $w \in k^4$ in the kernel of the linear map D_F. Euler's identity gives $0 = (D_w F)(w) = 3F(w)$, so $[w] \in X(k)$.

If $\operatorname{char} k = 3$, the hypothesis on $[L:k]$ implies that L/k is separable. Normalize v so that its i_0-th coordinate is 1 for some fixed i_0. Every conjugate $\sigma(v)$ satisfies (6.1), and directions satisfying this identity are closed under addition, since translations compose. Thus

$$w \coloneqq \operatorname{Tr}_{L/k}(v) = \sum_{\sigma : L \hookrightarrow \overline{k}} \sigma(v) \in k^4$$

also satisfies (6.1). By our normalization, the i_0-th coordinate of w is $[L:k] \neq 0$ in k, so $w \neq 0$. Setting $x = 0$ and $s = 1$ in (6.1) then gives $F(w) = 0$. This concludes the argument. □

Proposition 6.3. *Suppose every smooth cubic surface over $k((t))$ with a zero-cycle of degree one has a $k((t))$-rational point. Then every cubic surface over k with a zero-cycle of degree one has a k-rational point.*

Proof. Let $Y = \{F = 0\}$ be a cubic surface over k with a zero-cycle of degree one. Choose a closed point of degree d prime to three from the support of the given zero-cycle, and let L/k be its residue field extension. By Lemma 6.2, we may assume that there is a point $q \in Y_{\mathrm{sm}}(L)$.

Choose a homogeneous cubic G defining a smooth surface over k. For example, one may take

$$G \coloneqq \begin{cases} x_0^3 + x_1^3 + x_2^3 + x_3^3, & \operatorname{char} k \neq 3, \\ x_0^3 + x_0 x_1^2 + x_1 x_2^2 + x_2 x_3^2, & \operatorname{char} k = 3. \end{cases}$$

Put $R = k[[t]]$, $K = k((t))$, and

$$\mathscr{X} \coloneqq \{(1-t)F + tG = 0\} \subset \mathbb{P}^3_R.$$

This is a flat projective family with special fiber Y and smooth generic fiber $X \coloneqq \mathscr{X}_K$. Indeed, the pencil over $\mathbb{A}^1_k$ contains the smooth member $\{G = 0\}$ at $t = 1$, so its generic fiber is smooth.

The point q lifts to a point $\tilde{q} \in \mathscr{X}(L[[t]])$ by Hensel's lemma: on an affine chart containing q, choose a coordinate for which the partial derivative of the defining equation is nonzero at q. Fixing the other coordinates at their values at q, Hensel's lemma lifts the remaining coordinate to a power series in t satisfying the equation of $\mathscr{X}$; see also [Ma21, Lemma 2.4].

Put $M = L((t))$. Restricting $\tilde{q}$ to the generic point of $\operatorname{Spec} L[[t]]$ gives a point $\tilde{q}_\eta \in X(M)$. A k-basis of L is also a K-basis of M, so $[M : K] = d$. Let

$$\pi\colon X_M = X \times_K M \longrightarrow X$$

be the projection, and regard $\tilde{q}_\eta$ as an M-rational point of X_M. If P denotes its image in X, then

$$\zeta \coloneqq \pi_*[\tilde{q}_\eta] = [M : \kappa(P)] \cdot [P] \in \mathrm{CH}_0(X), \qquad \deg_K \zeta = d.$$

Since $\gcd(d, 3) = 1$, an integral linear combination of ζ and the degree-three intersection class with a line gives a zero-cycle of degree one on X.

Since X is smooth and has a zero-cycle of degree one, the hypothesis of the proposition implies that $X(K) \neq \varnothing$. By the valuative criterion of properness, any point of $X(K)$ extends to an R-point of $\mathscr{X}$, whose reduction belongs to $Y(k)$. □

Proof of Theorem 6.1. Apply Proposition 6.3, using Theorem 5.3 over $k((t))$. □

AI disclosure

The starting points for this paper were two different GPT-assisted proofs of the Cassels–Swinnerton-Dyer conjecture for cubic surfaces, obtained independently by the authors. The paper is the synthesis of these arguments, with GPT used to help with the copyediting. The authors' interactions with GPT and their respective contributions were as follows.

VA asked GPT-5.5 to find a counterexample to the conjecture, suggesting many ideas for it to try, including the universal family of four-pointed cubic surfaces as a candidate. GPT-5.5 made no progress, but when VA put the same question to GPT-6 Astra, it produced a rational section of this family. While the initial argument seemed convoluted, further exchanges with GPT led to the key idea presented in Section 3. At VA's request, GPT then developed extensions to arbitrary fields and arbitrary cubic forms, leading to the arguments presented in Sections 5 and 6.

Having set aside the search for counterexamples some time earlier, SS explored various approaches to constructing a rational point from a point of degree 4 in conversations with GPT. After exploring one such approach, involving quadrics through effective zero-cycles of degree 8 (cf. Section 4.1), GPT pointed out that the dimension of the space of quadrics through a nonreduced subscheme of length 8 supported at four general geometric points can jump for special choices of the tangent directions, cf. Section 4.2. It suggested that this observation offered a promising approach, while noting that the details still needed to be worked out and checked. SS developed and checked these details with GPT's assistance, leading to the results presented in Sections 4 and 5.

Acknowledgments

SS first learned about this problem from Johan de Jong in 2020 in connection with the work of his student Qixiao Ma, who subsequently held a postdoctoral position in SS's group from 2021 to 2023. VA learned about the problem from a talk given by Claire Voisin at a conference in Princeton in June of 2026.

We thank Jean-Louis Colliot-Thélène, Philip Engel, Qixiao Ma, Matthias Paulsen, Alexei Skorobogatov and Claire Voisin for discussions and comments.

VA was partially supported by NSF grant DMS-2501855. VA's free subscription to GPT was provided by the OpenAI for Academic Researchers program.

This project has also received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme under grant agreement No 948066 (ERC-StG RationAlgic). The research was partly conducted in the framework of the DFG-funded research training group RTG 2965: From Geometry to Numbers, Project number 512730679.

References

[AB18] A. Auel and M. Bernardara, *Semiorthogonal decompositions and birational geometry of del Pezzo surfaces over arbitrary fields*, Proc. Lond. Math. Soc. (3) **117** (2018), no. 1, 1–64.

[Bal24] F. Balestrieri, *Degrees of closed points on hypersurfaces*, Math. Nachr. **297** (2024), no. 5, 1831–1837.

[Beau00] A. Beauville, *Determinantal hypersurfaces*, Michigan Mathematical Journal **48** (2000), 39–64.

[BN18] A. Bremner and T. N. Xuan, *An interesting quartic surface, everywhere locally solvable, with cubic point but no global point*, Publ. Math. Debrecen **93** (2018), no. 1–2, 253–260.

[BE77] D. A. Buchsbaum and D. Eisenbud, *Algebra structures for finite free resolutions, and some structure theorems for ideals of codimension* 3, American Journal of Mathematics **99** (1977), no. 3, 447–485.

[Cor76a] D. F. Coray, *Algebraic points on cubic hypersurfaces*, Acta Arith. **30** (1976), no. 3, 267–296.

[Cor76b] D. F. Coray, *Arithmetic on singular cubic surfaces*, Compositio Mathematica **33** (1976), no. 1, 55–67.

[Cor77] D. F. Coray, *Points algébriques sur les surfaces de del Pezzo*, C. R. Acad. Sci. Paris Sér. A-B **284** (1977), no. 24, 1531–1534.

[CT20] J.-L. Colliot-Thélène, *Zéro-cycles sur les surfaces de del Pezzo (Variations sur un thème de Daniel Coray)*, Enseign. Math. (2) **66** (2020), no. 3–4, 447–487.

[CTC79] J.-L. Colliot-Thélène and D. Coray, *L'équivalence rationnelle sur les points fermés des surfaces rationnelles fibrées en coniques*, Compositio Math. **39** (1979), no. 3, 301–332.

[CTSD12] J.-L. Colliot-Thélène and P. Swinnerton-Dyer, *Zero-cycles and rational points on some surfaces over a global function field*, Acta Arith. **155** (2012), 63–70.

[CTW26] J.-L. Colliot-Thélène and O. Wittenberg, *Principe de Hasse pour les hypersurfaces cubiques lisses dans le cas fonctionnel*, Preprint 2026.

[Eis05] D. Eisenbud, *The Geometry of Syzygies: A Second Course in Algebraic Geometry and Commutative Algebra*, Graduate Texts in Mathematics **229**, Springer, New York, 2005.

[EKM08] R. Elman, N. Karpenko, and A. Merkurjev, *The algebraic and geometric theory of quadratic forms*, American Mathematical Society Colloquium Publications, vol. 56, American Mathematical Society, Providence, RI, 2008.

[Fog68] J. Fogarty, *Algebraic families on an algebraic surface*, American Journal of Mathematics **90** (1968), no. 2, 511–521.

[Han14] F. Han, *Pfaffian bundles on cubic surfaces and configurations of planes*, Mathematische Zeitschrift **278** (2014), nos. 1–2, 363–383.

[Kol02] J. Kollár, *Unirationality of cubic hypersurfaces*, J. Inst. Math. Jussieu **1** (2002), no. 3, 467–476.

[KM17] J. Kollár and M. Mella, Quadratic families of elliptic curves and unirationality of degree 1 conic bundles, Amer. J. Math. **139** (2017), no. 4, 915–936.

[Ma21] Q. Ma, *Closed points on cubic hypersurfaces*, Michigan Math. J. **70** (2021), no. 4, 857–868.

[Man86] Yu. I. Manin, *Cubic forms: algebra, geometry, arithmetic*, second edition, North-Holland Mathematical Library, vol. 4, North-Holland, Amsterdam, 1986.

[Poi01] H. Poincaré, *Sur les propriétés arithmétiques des courbes algébriques*, J. Math. Pures Appl. (5) **7** (1901), 161–233.

[Por25] M. Porzio, *The arithmetic of del Pezzo surfaces and Hilbert schemes of points*, Ph.D. thesis, Columbia University, 2025.

[RV22] C. Rivera and B. Viray, *Persistence of the Brauer–Manin obstruction on cubic surfaces*, Math. Res. Lett. **29** (2022), no. 6, 1881–1889.

[Seg51] B. Segre, *On the rational solutions of homogeneous cubic equations in four variables*, Math. Notae **11** (1951), 1–68.

[Spr52] T. A. Springer, *Sur les formes quadratiques d'indice zéro*, C. R. Acad. Sci. Paris **234** (1952), 1517–1519.

[Stacks] The Stacks Project Authors, *The Stacks Project*, https://stacks.math.columbia.edu/, 2026.

[Tan14] F. Tanturri, *Pfaffian representations of cubic surfaces*, Geometriae Dedicata **168** (2014), 69–86.

[Voi26] C. Voisin, *Rank 2 vector bundles and degrees of points of del Pezzo surfaces*, J. Reine Angew. Math. **836** (2026), 303–327. (bicentennial issue 2026)

Department of Mathematics, University of Georgia, Athens, GA 30602, USA
Email address: valery@math.uga.edu

Leibniz University Hannover, Institute of Algebraic Geometry, Welfengarten 1, 30167 Hannover, Germany
Email address: schreieder@math.uni-hannover.de